We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**
Mefentrifluconazole 400 G/L SC

Use: fungicide

Manufacturer/supplier:
BASF South East Asia Pte Ltd.
128 Beach Road #18-01
Guoco Midtown, 189773, Singapore
Telephone: +65 8322 4420
Telefax number: +65 6 334-0330
E-mail address: benny.zou@basf.com

Emergency information:
Singapore Emergency Toll-Free Number:
Telephone: 1800-723-1361
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazard identification

Classification of the substance and mixture:
Hazardous to the aquatic environment - acute: Cat.2
Hazardous to the aquatic environment - chronic: Cat.1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Warning

Hazard Statement:

| | |
|---|---|
| H401 | Toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Response):

| | |
|---|---|
| P391 | Collect spillage. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

May produce an allergic reaction. Contains:
(2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole, 1,2-benzisothiazol-3(2H)-one, 2-methyl-2H-isothiazol-3-one

## 3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, fungicide

**Hazardous ingredients**

1H-1,2,4-Triazole-1-ethanol, α-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-α-methyl-

| | |
|---|---|
| Content (W/W): 34.93 % | Skin Sens.: Cat. 1 |
| CAS Number: 1417782-03-6 | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |
| | M-factor acute: 1 |
| | M-factor chronic: 1 |

Benzenesulfonic acid, hydroxy-, polymer with formaldehyde, phenol and urea, sodium salt

| | |
|---|---|
| Content (W/W): < 3 % | Eye Dam./Irrit.: Cat. 2A |
| CAS Number: 102980-04-1 | Aquatic Acute: Cat. 3 |
| | Aquatic Chronic: Cat. 3 |

Residues (petroleum), catalytic reformer fractionator, sulfonated,polymers with formaldehyde, sodium salts

| | |
|---|---|
| Content (W/W): < 3 %<br>CAS Number: 68425-94-5 | Eye Dam./Irrit.: Cat. 2A<br>Aquatic Acute: Cat. 3<br>Aquatic Chronic: Cat. 3 |

1,2-benzisothiazol-3(2H)-one

| | |
|---|---|
| Content (W/W): < 0.036 %<br>CAS Number: 2634-33-5 | Acute Tox.: Cat. 2 (Inhalation - dust)<br>Acute Tox.: Cat. 4 (oral)<br>Skin Irrit.: Cat. 2<br>Eye Dam.: Cat. 1<br>Skin Sens.: Cat. 1A<br>Aquatic Acute: Cat. 1<br>Aquatic Chronic: Cat. 1<br>M-factor acute: 1<br>M-factor chronic: 1 |

2-methyl-2H-isothiazol-3-one

| | |
|---|---|
| Content (W/W): < 0.01 %<br>CAS Number: 2682-20-4 | Acute Tox.: Cat. 2 (Inhalation - dust)<br>Acute Tox.: Cat. 3 (oral)<br>Acute Tox.: Cat. 3 (dermal)<br>Skin Corr.: Cat. 1B<br>Eye Dam.: Cat. 1<br>Skin Sens.: Cat. 1A<br>Aquatic Acute: Cat. 1<br>Aquatic Chronic: Cat. 1<br>M-factor acute: 10<br>M-factor chronic: 1 |

propane-1,2-diol

Content (W/W): < 10 %
CAS Number: 57-55-6

## 4. First-Aid Measures

General advice:
Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen chloride, hydrogen fluoride, nitrogen oxides, halogenated compounds, sulfur oxides, silica compounds
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray.Use personal protective clothing.Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into the subsoil/soil.Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 60 Months

Protect from temperatures below: -5 °C
Changes in the properties of the product may occur if substance/product is stored below indicated temperature for extended periods of time.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

## 8. Exposure controls and personal protection

Components with occupational exposure limits

1H-1,2,4-Triazole-1-ethanol, α-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-α-methyl-, 1417782-03-6;
TWA value 0.68 mg/m3 (BASF recomm. occupational exposure limit)

Personal protective equipment

Respiratory protection:
Respiratory protection not required.

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

| | |
|---|---|
| Form: | liquid |
| Colour: | cream |
| Odour: | smoky, faintly aromatic |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |
| pH value: | approx. 6 - 8<br>(20 °C) |
| Melting point: | approx. 0 °C<br>Information applies to the solvent. |
| boiling temperature: | approx. 95 °C |
| Flash point: | No flash point - Measurement made up to the boiling point. |
| Evaporation rate: | No data available. |
| Flammability (solid/gas): | not flammable |
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |

Ignition temperature: approx. 652 °C

Thermal decomposition: 160 °C , 20 kJ/kg
(onset temperature)

300 °C , 90 kJ/kg
(onset temperature)

Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.

SADT: No data available.

Explosion hazard: not explosive

Fire promoting properties: not fire-propagating

Vapour pressure: approx. 23.4 hPa
(20 °C)
Information applies to the solvent.

Density: approx. 1.15 g/cm3
(20 °C)
approx. 1.14 g/cm3
(50 °C)

Relative density:
No data available.

Relative vapour density (air):
not determined

Solubility in water: dispersible

Partitioning coefficient n-octanol/water (log Pow):
not applicable for mixtures

Viscosity, dynamic: approx. 76 mPa.s
(20 °C, 100 1/s)

Viscosity, kinematic: approx. 60 mm2/s
(40 °C)

No data available.

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

Specific Surface Area:
No data available.

Particle Shape:
No data available.

Dustiness:
No data available.

## 10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

| | |
|---|---|
| Thermal decomposition: | 160 °C, 20 kJ/kg<br>(onset temperature) |
| Thermal decomposition: | 300 °C, 90 kJ/kg<br>(onset temperature) |
| Thermal decomposition: | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. |

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

**Acute oral toxicity**

Experimental/calculated data:
LD50rat (oral): > 2,000 mg/kg
No mortality was observed.

**Acute inhalation toxicity**

LC50 rat (by inhalation): > 5.48 mg/l
No mortality was observed. An aerosol was tested.

**Acute dermal toxicity**

LD50 rat (dermal): > 5,000 mg/kg
No mortality was observed.

**Assessment of acute toxicity**

Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

### Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit: non-irritant

### Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Experimental/calculated data:
Mouse Local Lymph Node Assay (LLNA) mouse: Non-sensitizing.

### Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

### Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of various animal studies gave no indication of a carcinogenic effect.

Experimental/calculated data:
No data available.

### Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Experimental/calculated data:
No data available.

### Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

### Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Experimental/calculated data:
No data available.

Information on: (2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole
Assessment of repeated dose toxicity:
Repeated oral exposure to large quantities may affect certain organs. Liver Based on available data, the classification criteria are not met.
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### Aspiration hazard

not applicable

### Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Toxic to aquatic life. Very toxic to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) 1.44 mg/l, Oncorhynchus mykiss (OECD Guideline 203)

Aquatic invertebrates:
EC50 (48 h) 3.4 mg/l, Daphnia magna (OECD Guideline 202, part 1)

Aquatic plants:
EC50 (72 h) 8.360 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

EC10 (72 h) 1.422 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

Microorganisms/Effect on activated sludge:
No data available.

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:
No data available.

Information on: (2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole
Chronic toxicity to fish:
No observed effect concentration (36 d) 0.027 mg/l, Brachydanio rerio
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Information on: (2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.01 mg/l, Daphnia magna

EC10 (21 d), 0.0175 mg/l, Daphnia magna
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## Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: (2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.
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## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: (2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole
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## Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: (2RS)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl)propan-2-ol; mefentrifluconazole
Bioaccumulation potential:
Bioconcentration factor: 385
Does not accumulate in organisms.
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### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (MEFENTRIFLUCONAZOLE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Sea transport**
IMDG

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (MEFENTRIFLUCONAZOLE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-F |

**Air transport**
IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (MEFENTRIFLUCONAZOLE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

**Further information**

Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

## 15. Regulatory Information

**Other regulations**

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.